Code No: 134AP **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February -2024
DATABASE MANAGEMENT SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Define logical data independence.	[2]
b)	What is a view?	[3]
c)	What is procedural DML?	[2]
d)	What is an active database?	[3]
e)	What are spurious tuples?	[2]
f)	What is transitive rule for functional dependency?	[3]
g)	Define serializable schedule.	[2]
h)	What is the motivation for concurrent transactions?	[3]
i)	What is the use of cylinder on a disk?	[2]
j)	List the drawbacks of ISAM indexing technique.	[3]

PART – B

(50 Marks)

2. Explain the role of ER model in database design process with suitable examples. [10]

OR

3. What is an integrity constraint? Explain enforcement of various integrity constraints in DBMS. [10]

4. With suitable example expressions, illustrate various operators available in Relational Algebra. [10]

OR

5. Consider the following database schema to write queries in SQL.
Student(Rollno, name, totalCredits)
Course(Cno, cname, credits)
Enroll(Rollno, Cno, Grade)
a) Find students who scored grade A in DBMS
b) Find the students enrolled for DBMS and CO
c) Find the courses having enrollment less than 50. [3+3+4]

6. What is redundancy? Explain the problems caused by redundancy. [10]

OR

7. Does third normal form ensure dependency preservation? Justify your answer with suitable examples. [10]

8. Explain how time stamp based protocol ensures serializability of schedules. [10]

OR

9. Demonstrate buffer management to support recovery techniques. [10]

10. Describe the significance of multi level indexing and its influence on tree based indexing. [10]

OR

11. Make a comparison of extendible hashing and linear hashing techniques as indexing structures. [10]

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